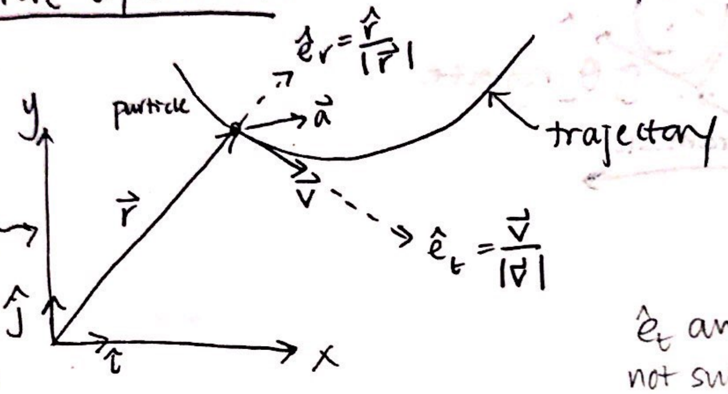
Particle dynamics:



capital letter F that indicated fixed (Newtonian frame)

Particle
- All you care about is average motion
  - Ignore distortion + rotation
  - Pay attention to center of mass motion

$\hat{e}_t$ and $\hat{e}_r$ not supposed to be $\perp$ ("Murphy's Law of Drawing")

Problem: **FBD:**

all external forces on particle

$$\vec{F} = \vec{F}(\vec{r}, \vec{v}, \vec{t})$$

LMB: $\vec{F}_{tot} = m_{tot}\,\vec{a}$ — a of C.O.M.

E.O.M:

$$\boxed{\begin{aligned}\dot{\vec{r}} &= \vec{v}\\ \dot{\vec{v}} &= \frac{\vec{F}(\vec{r},\vec{v},\vec{t})}{m}\end{aligned}}$$

· At an instant, r and v are independent

How many 1st order ODEs.

1D	2D	3D
2	4	6

$$\Rightarrow \dot{z} = f(t, z)$$

$$\underbrace{z = \begin{bmatrix} x \\ y \\ z \\ v_x \\ v_y \\ v_z \end{bmatrix}}_{\text{state}}$$

3 Pillars

I. Force Laws:
- How does $\vec{F}$ depend on $\vec{r}, \vec{v}, \vec{t}$? $\quad \vec{F}(r, \vec{v}, t)$
- Model of forces
- constitutive laws

II. Geometry in Motion
e.g. $\dot{\vec{r}} = \vec{v}$, $\dot{\vec{v}} = \vec{a}$

III. Laws of Mechanics
$\vec{F} = m\vec{a}$

Examples | Ballistics:

cannonball

actual trajectory, not really parabolic

FBD:

$$\vec{F}_D = \begin{cases} \vec{0} & \text{no drag} \\ -c\vec{v} & \text{linear drag} \\ -c|\vec{v}|\vec{v} = -c\vec{v}^2\hat{e}_t & \text{quadratic drag} \end{cases}$$

if controlling force, then t becomes relevant

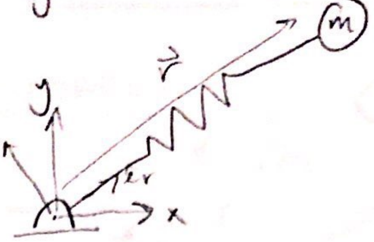
Computer

```
f = @(t,z)      myrhs(t,z,p)
[tarray, zarray] = ode45(f, .....)

function zdot = myrhs(t,z,p)
   r= z(1:3)    % 3D version
   v= z(4:6)
   rdot = v
   vdot = mechanics
   zdot = [rdot; vdot]';
end
```

what this course covers

ex) Spring



FBD:

$\vec{F}_s = -k(|\vec{r}| - r_0)\, \underline{e}_r$ (?)

$-k\vec{r}$ (if $l_0 = 0$)

both correct

ex) Gravity.

$\vec{F}_{12} = \dfrac{m_1 m_2 G}{r^3}\, \vec{r}_{12}$

force on 1 from 2

$F = \dfrac{-mg Re^2 \vec{r}}{r^3}$ $\quad \dfrac{\hat{e}_r}{\vec{r}}$

ex) Electric charge:

$\vec{F}_{12} = c q_1 q_2$ electrostatics

ex) Magnetic

$\vec{F} = c\vec{b} \times \vec{V} q$

Theorems / Facts / Properties

Patterns in solution

Model + IC $\Rightarrow$ motion

Linear momentum: $\vec{L} \equiv m\vec{v}$

Angular momentum: $\vec{H}_{/o} \equiv \vec{r}_{/o} \times (m\vec{v})$

Kinetic energy: $E_K = \frac{1}{2}mv^2 = \frac{1}{2}m\vec{v}\cdot\vec{v}$